

compass-style.org/

- Bootstrap - <http://getbootstrap.com/css/>
- The dynamic stylesheet language - <http://lesscss.org/>
- 'CSS: The Definitive Guide', 3rd Edition, by Eric A Meyer, O'Reilly

Frameworks and patterns

As the saying goes, "A good programmer knows what to write and a great programmer knows what to rewrite." So it's unlikely that you'll be asked to code a login page over and over again, especially when frameworks and software patterns are available for free. Being aware about the frameworks available for your scripting language is essential and a sound knowledge of at least one of them is a must.

Browse the following to know more:

- Zend Framework - <http://www.zend.com/en/products/framework/>
- CodeIgniter: A fully baked PHP framework - <http://ellislab.com/codeigniter>
- PEAR – PHP extension and application repository - <http://pear.php.net/>
- Django: The Web framework for perfectionists with deadlines - <https://www.djangoproject.com/>
- The Pylons Project - <http://www.pylonsproject.org/>
- JQuery: A feature-rich JavaScript library - <http://jquery.com/>
- Dojo: <http://dojotoolkit.org/>
- Mootools: A compact Javascript framework - <http://mootools.net/>
- Ruby on Rails: Web development that doesn't hurt - <http://rubyonrails.org/>
- 'Zend Framework, A Beginner's Guide', 1st Edition, by Vikram Vaswani, McGraw-Hill
- 'Zend Framework In Action', by Rob Allen, Nick Lo and Steven Brown, Manning
- 'Professional CodeIgniter', by Thomas Myer, Wrox
- 'Dojo: The Definitive Guide', 1st Edition, by Matthew A Russell, O'Reilly

HTTP servers

Apache is the Web server of choice for hosting Web applications. Knowing how it works is essential. To truly exploit its powers, get your hands dirty on the Apache directives.

NGINX is gaining popularity as a low memory footprint HTTP server for PHP and Perl applications using fast CGI. It can also be deployed as a HTTP load balancer.

Lighttpd is a lightweight HTTP server usually used to serve static content and complement the main HTTP server so that page downloads to the browser can be accelerated.

Browse the following to know more:

- Apache HTTP server - <http://projects.apache.org/>

projects/http_server.html

- NGINX - <http://nginx.org/>
- Lighttpd - <http://www.lighttpd.net/>
- 'Apache: The Definitive Guide', 3rd Edition, by Ben Laurie and Peter Laurie, O'Reilly

Architecture

To build industrial strength software applications, you'll need a sound architecture represented with an appropriate notation like IDEFIX or UML. As a programmer, you would be required to understand the rules conveyed in these architecture diagrams, which need to be built into the code. So a sound knowledge of IDEFIX and UML is a must. Further knowledge of the software architectural patterns is definitely an advantage.

To know more, read:

- 'Design Patterns: Elements of Reusable Object-Oriented Software', 3rd Edition, by Eric Gamma, Richard Helm, Ralph Johnson and John Vlissides, Addison-Wesley

Supporting disciplines

Software development has supporting disciplines like testing and versioning of the various artefacts. So a sound knowledge of automated testing and version control is essential to be a great software developer. Debugging is another skill that you should cultivate to fix your applications, so that they work the way they should.

Browse the following to know more:

- Concurrent Versions System - <http://www.nongnu.org/cvs/>
- Apache Subversion - <http://subversion.apache.org/>
- Git - <http://git-scm.com/>
- PHPUnit - <http://phpunit.sourceforge.net/>
- Firebug - <https://getfirebug.com/>
- Webdeveloper Toolbar - <http://chrispederick.com/work/web-developer/>

Regular expressions

Regular expressions (regex) are used extensively in programming to find patterns for replacement. In your career as a programmer, you will use regular expressions extensively in your programming and research. Most HTTP server redirect rules are based on regular expressions. For that matter, you may have used regular expressions unknowingly. Surprised? *,*, which is the notation for a file of any name and any extension, is the simplest example to illustrate the power of regex. So get yourself acquainted with regex for a truly fruitful experience.

Browse through:

- 'Mastering Regular Expressions', 3rd Edition, by Jeffrey E F Friedl, O'Reilly